



ESAME:

Grafo diretto o non diretto
e quasi sempre pesato

Intro to Graphs

Theoretical foundations

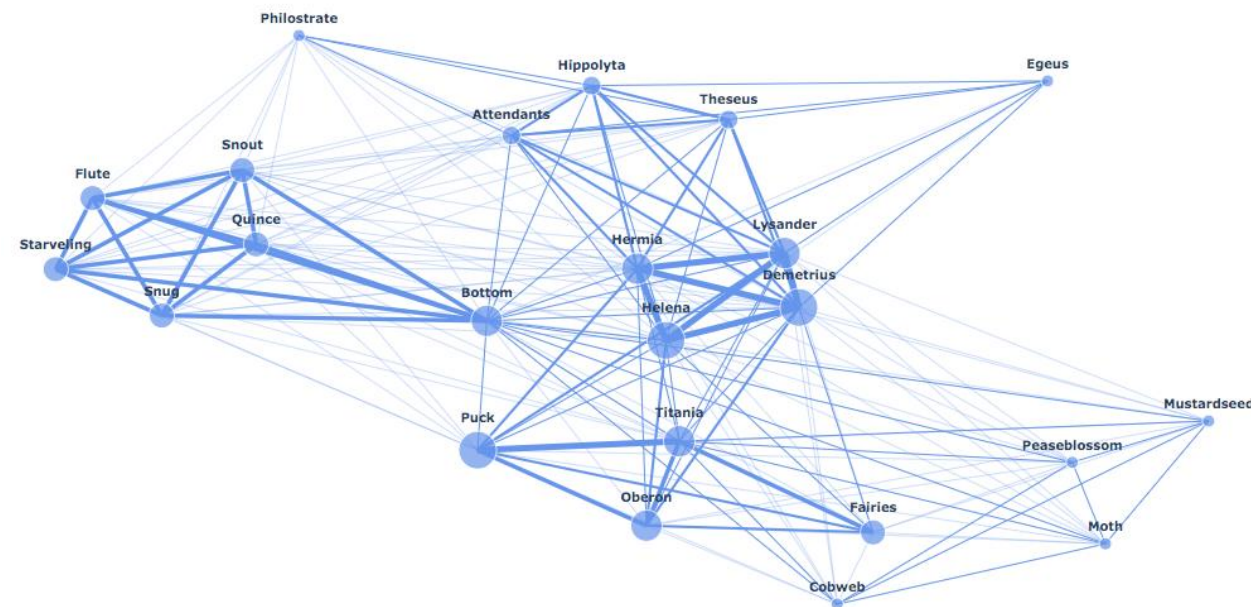
Giuseppe Averta

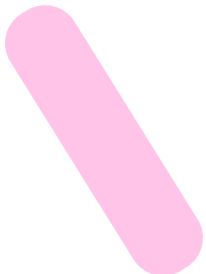
Carlo Masone

Francesca Pistilli

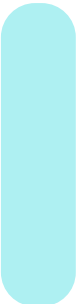
Davide Buoso

Gaetano Falco





Introduction to Graphs

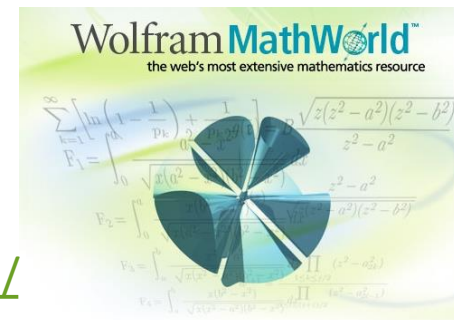


DEFINITION: GRAPH

Definition: Graph

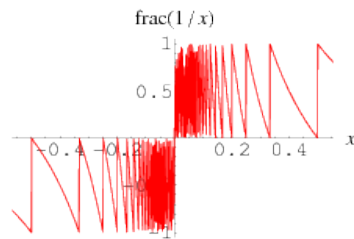
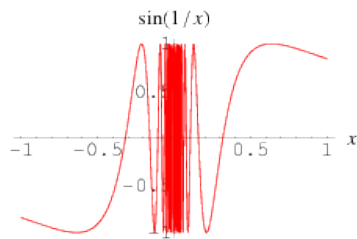
- A **graph** is a collection of **points** and **lines** connecting some (eventually empty) subset of them.
- The points of a graph are typically known as **graph vertices**, but may also be called “nodes” or simply “points.”
- The lines connecting the vertices of a graph are called **graph edges**, but may also be called “arcs” or “lines.”

<http://mathworld.wolfram.com/>

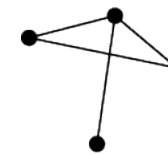


Big warning: Graph $\neq$ Graph $\neq$ Graph

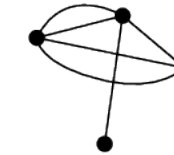
- Graph (plot)
- (italiano: grafico)



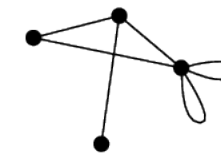
- Graph (maths)
- (*italiano: grafo*)



simple graph



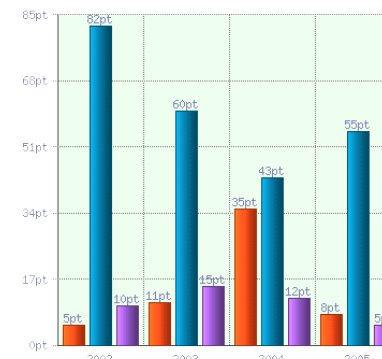
nonsimple graph
with multiple edges



nonsimple graph
with loops

$\neq$

Graph (chart)
(italiano: grafico)



History

- The study of graphs is known as **graph theory**, and was first systematically investigated by D. König in the 1930s
- Euler's proof about the *walk across all seven bridges of Königsberg* (1736), now known as the *Königsberg bridge* problem, is a famous precursor to graph theory.
- Indeed, the study of various sorts of paths in graphs has many applications in real-world problems.

Königsberg Bridge Problem

- Can the 7 bridges of the city of Königsberg over the river Pregel all be traversed in a single trip without doubling back, with the additional requirement that the trip ends in the same place it began?

1 solo passaggio a ponte

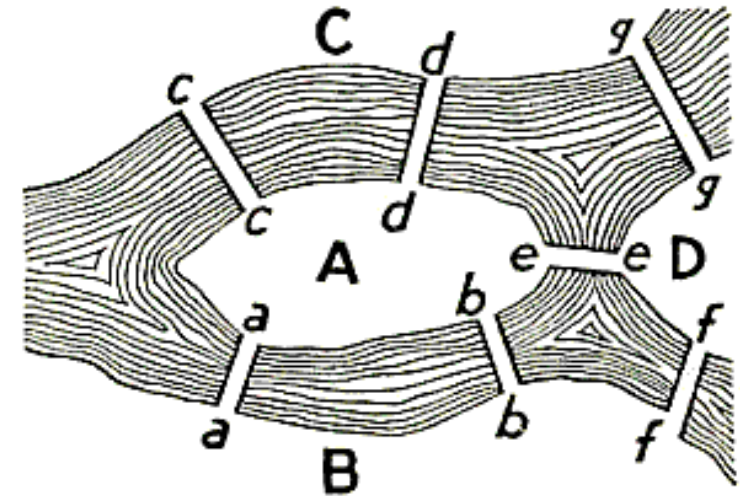
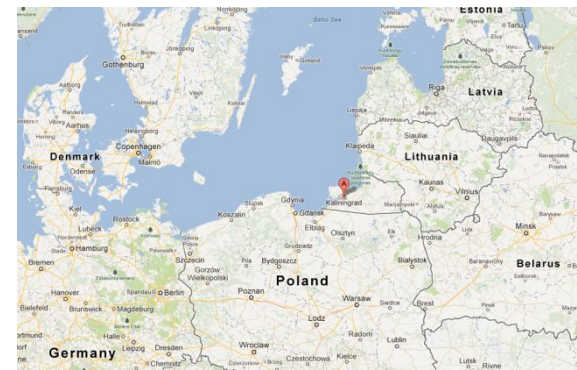


FIGURE 98. *Geographic Map:
The Königsberg Bridges.*



Today: Kaliningrad, Russia

Königsberg Bridge Problem

- Can the 7 bridges of the city of Königsberg over the river Pregel all be traversed in a single trip without doubling back, with the requirement that the trip ends where it began?

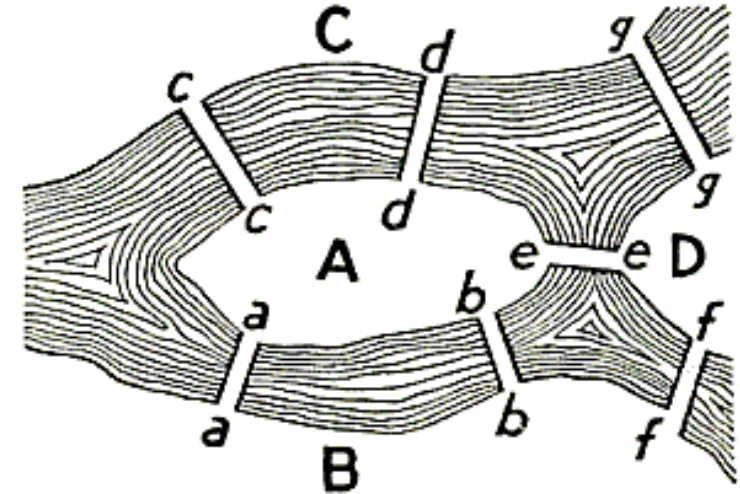
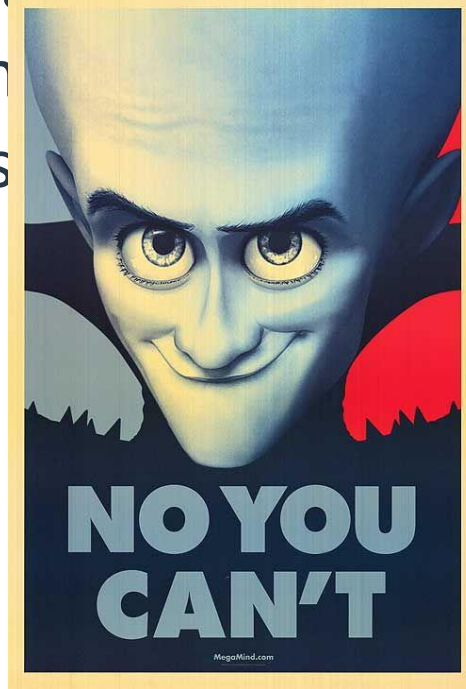
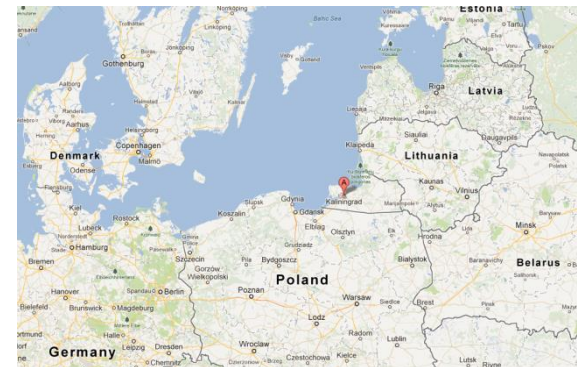


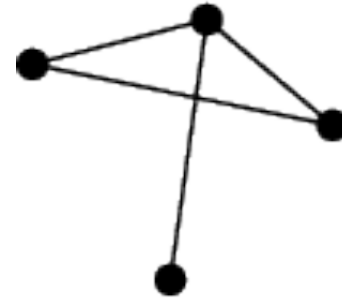
FIGURE 98. *Geographic Map:
The Königsberg Bridges.*



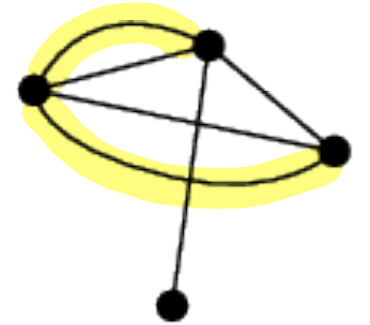
Today: Kaliningrad, Russia

Types of graphs: edge cardinality

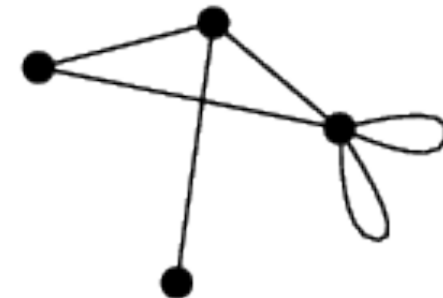
- Simple graph:
 - At most one edge (i.e., either one edge or no edges) may connect any two vertices
- Multigraph:
 - Multiple edges are allowed between vertices
- Loops:
 - Edge between a vertex and itself
- Pseudograph:
 - Multigraph with loops



simple graph



multigraph



loop

pseudograph

Types of graphs: edge direction

- Undirected
- Oriented
 - Edges have one direction (indicated by arrow)
- Directed
 - Edges may have one or two directions
- Network
 - Oriented graph with weighted edges

connesso → ogni nodo raggiungibile (direttamente o indirettamente)

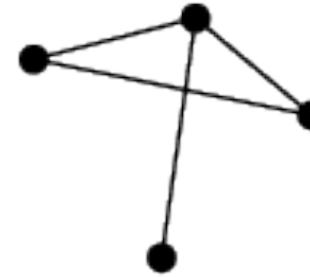
completo → ogni arco connesso a tutti gli altri

densità → $\frac{\# \text{ archi}}{\# \text{ archi possibili}}$

indiretta e aciclica

↓
foresta

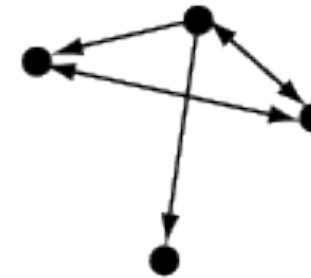
" e " e connesso
↓
albero



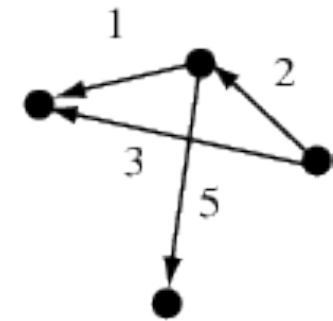
undirected graph



oriented graph



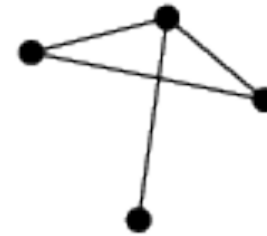
directed graph



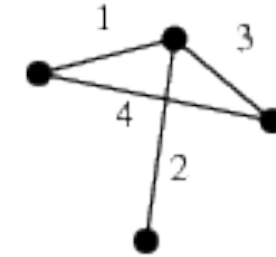
network

Types of graphs: labeling

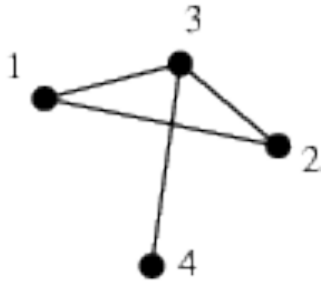
- Labels
 - None
 - On Vertices
 - On Edges
- Groups (=colors)
 - Of Vertices
 - no edge connects two identically colored vertices
 - Of Edges
 - adjacent edges must receive different colors
 - Of both



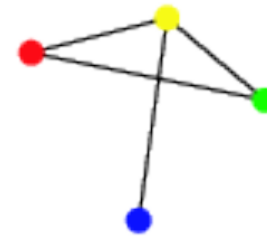
unlabeled graph



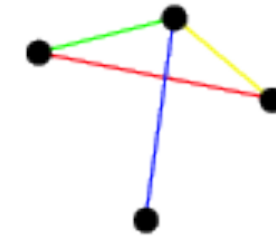
edge-labeled graph



vertex-labeled graph



vertex-colored graph



edge-colored graph

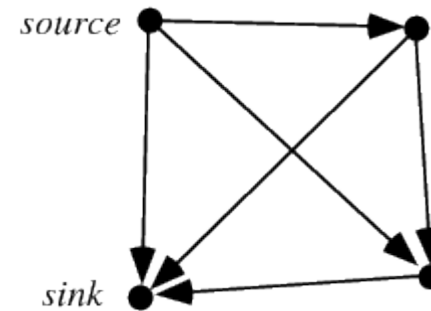


vertex- and edge-colored graph

Directed and Oriented graphs

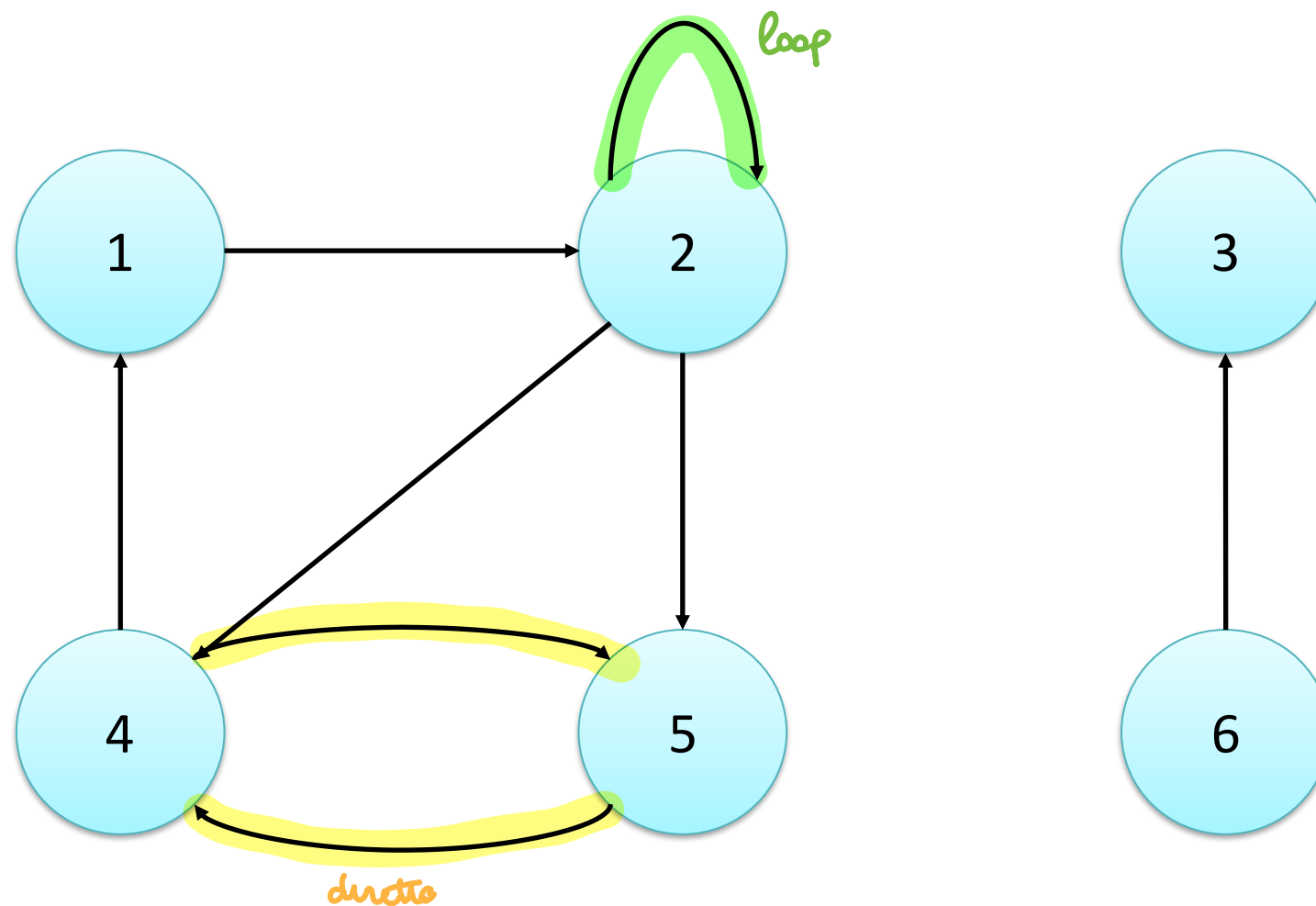
- A Directed Graph (*di-graph*) G is a pair (V, E) , where
 - V is a (finite) set of *vertices*
 - E is a (finite) set of *edges*, that identify a binary relationship over V
 - $E \subseteq V \times V$

NON
ORDINATI !

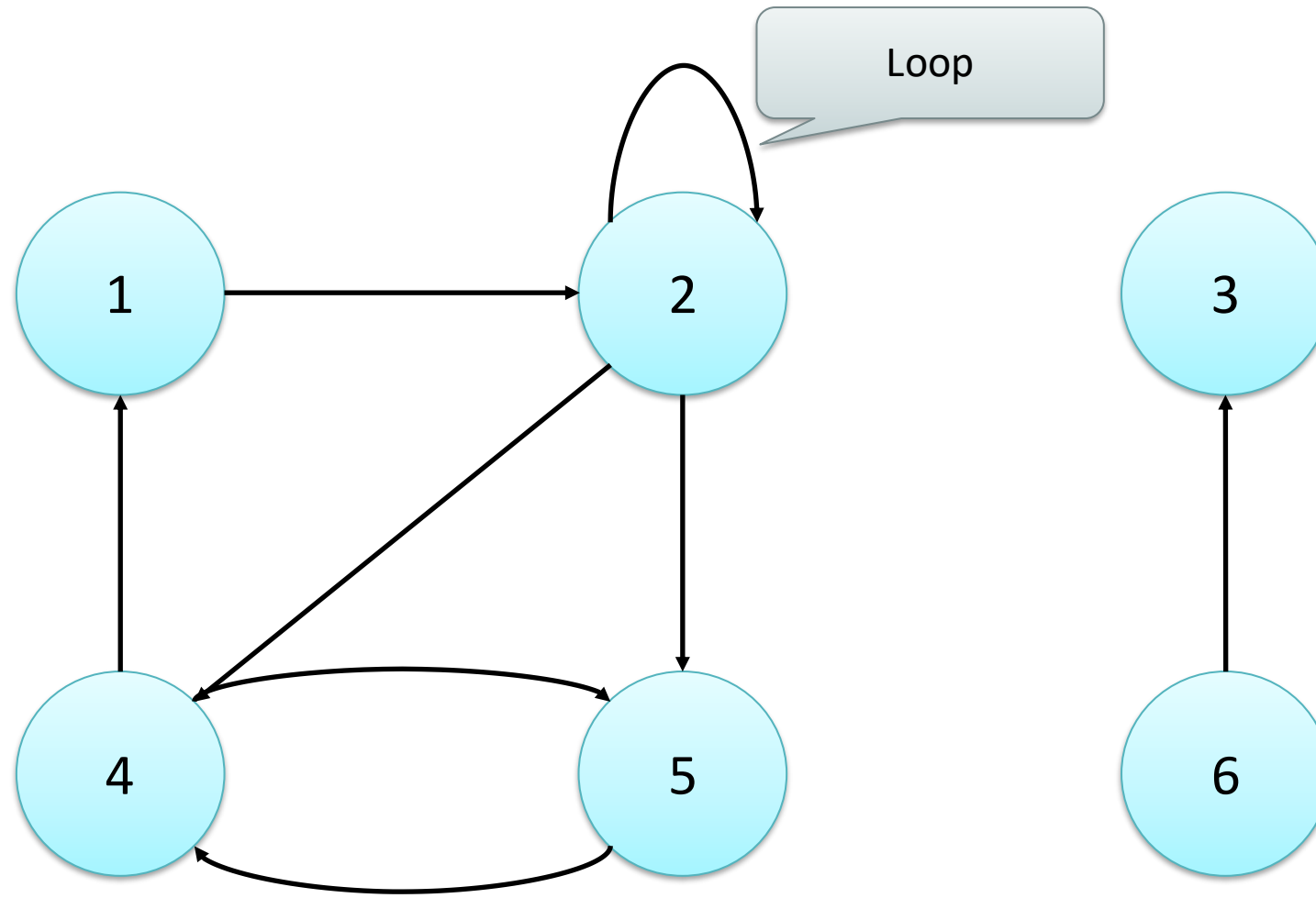


Example

Grafo unico NON connesso



Example

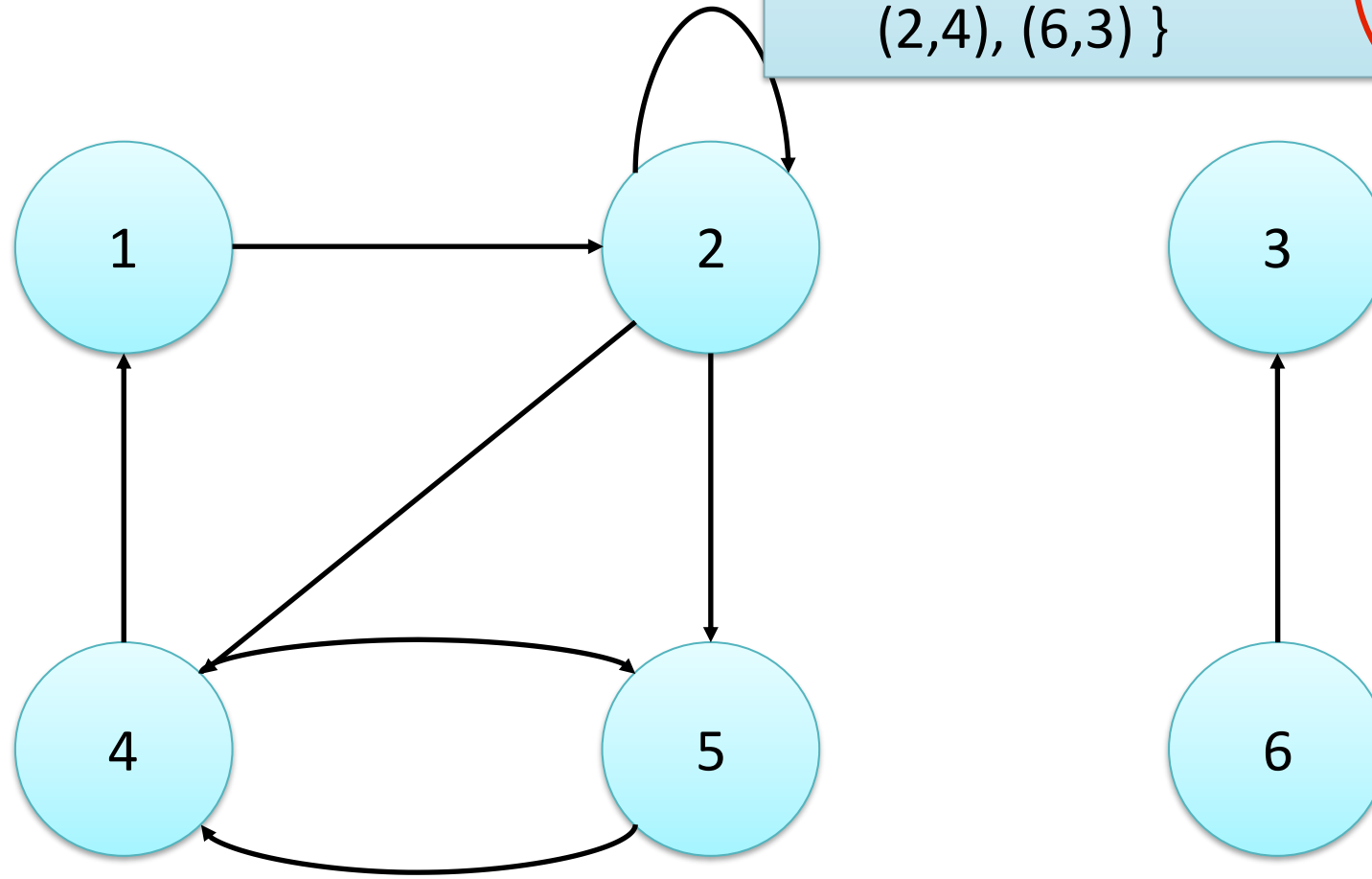


Example

$V = \{1,2,3,4,5,6\}$

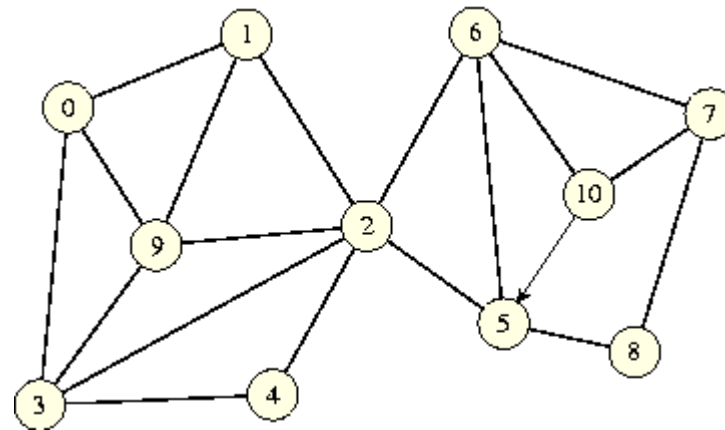
$E = \{ (1,2), (2,2), (2,5), (5,4), (4,5), (4,1), (2,4), (6,3) \}$

IMPORTANTE
L'ORDINE



Undirected graph

- An **Undirected** Graph is still represented as a tuple $G=(V,E)$, but the set E is made of **non-ordered pairs** of vertices

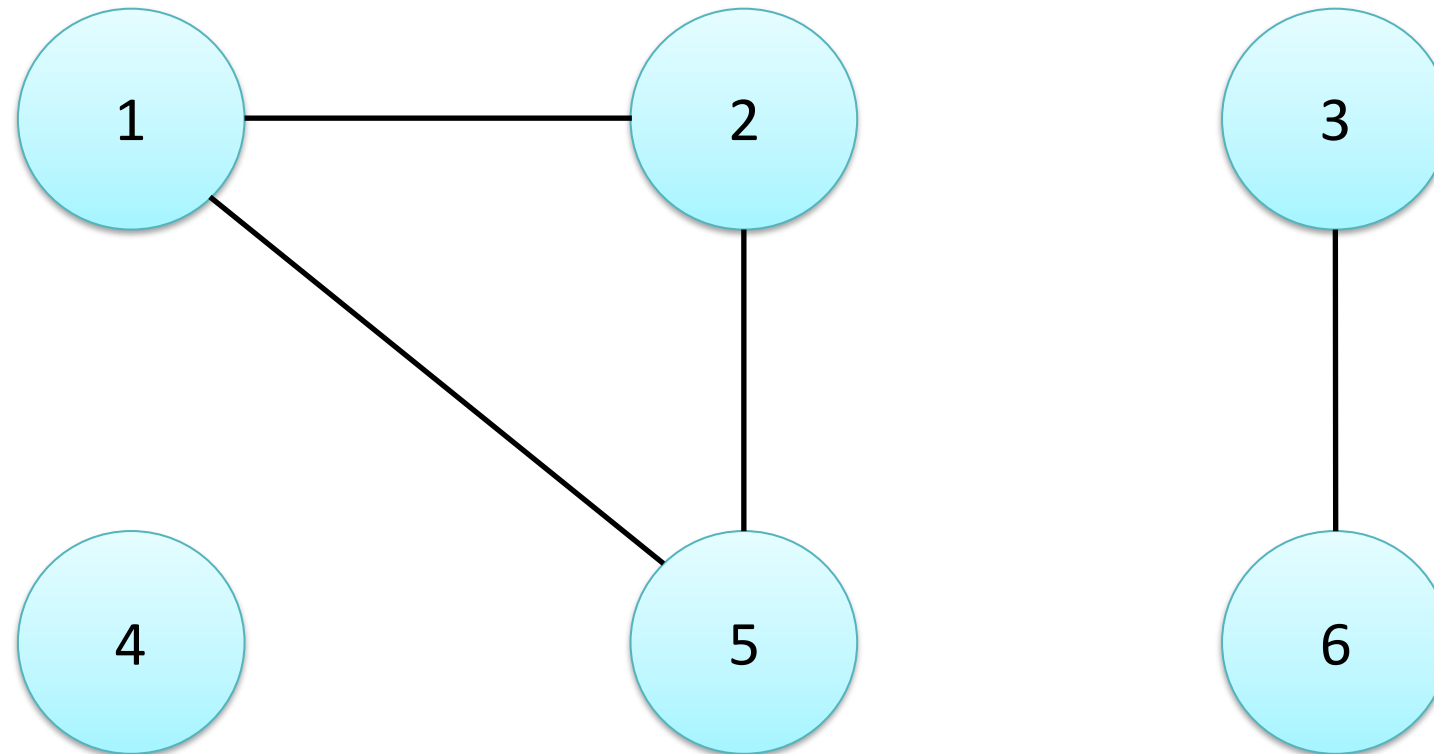


Example

$V = \{ 1, 2, 3, 4, 5, 6 \}$

$E = \{ \{1,2\}, \{2,5\}, \{5,1\}, \{6,3\} \}$

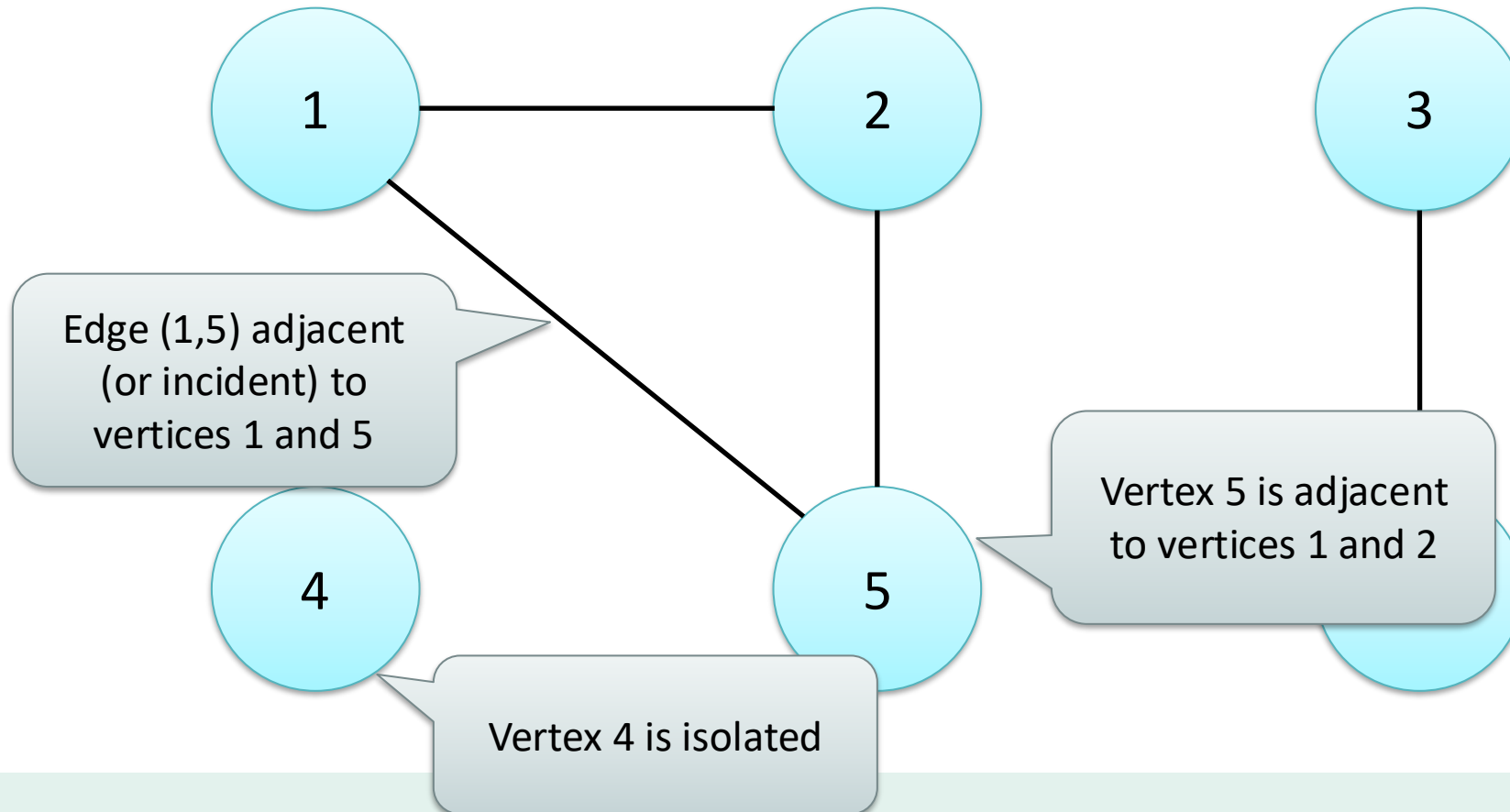
*non
importe
ordine*

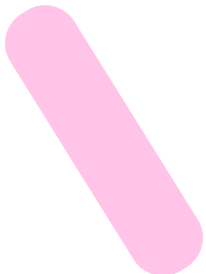


Example

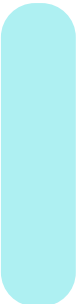
$V = \{ 1, 2, 3, 4, 5, 6 \}$

$E = \{ \{1,2\}, \{2,5\}, \{5,1\}, \{6,3\} \}$





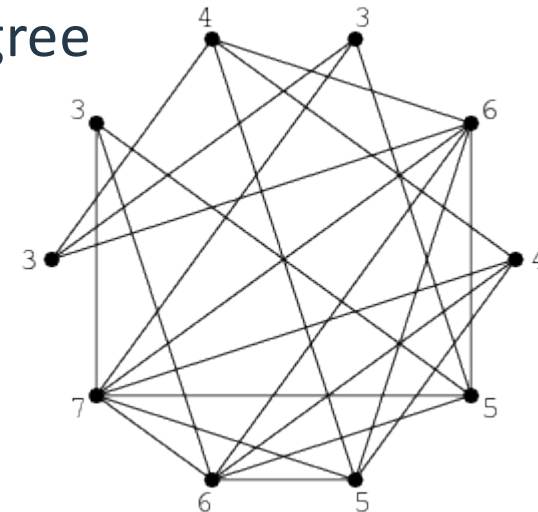
Introduction to Graphs



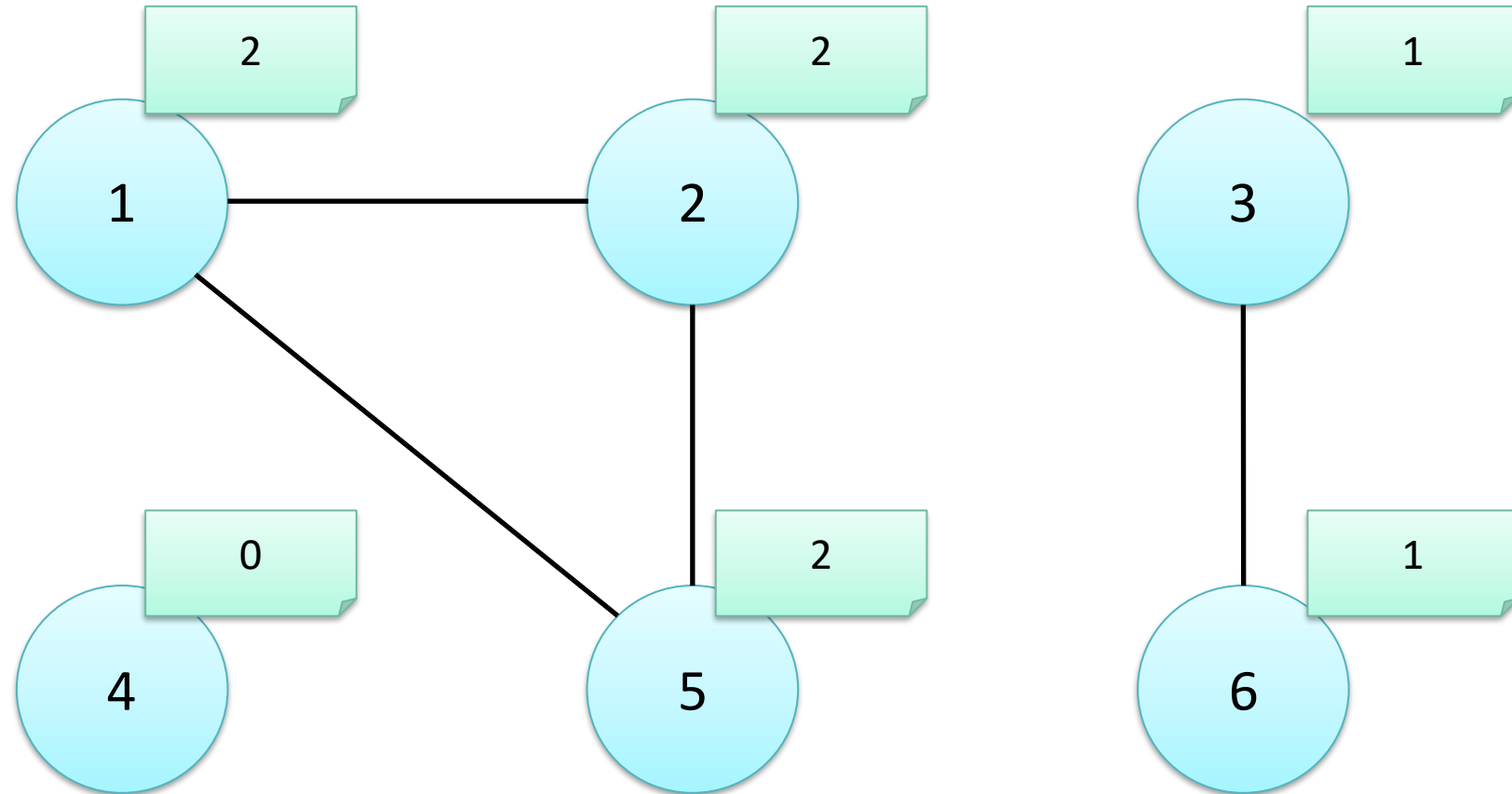
RELATED DEFINITIONS

Degree

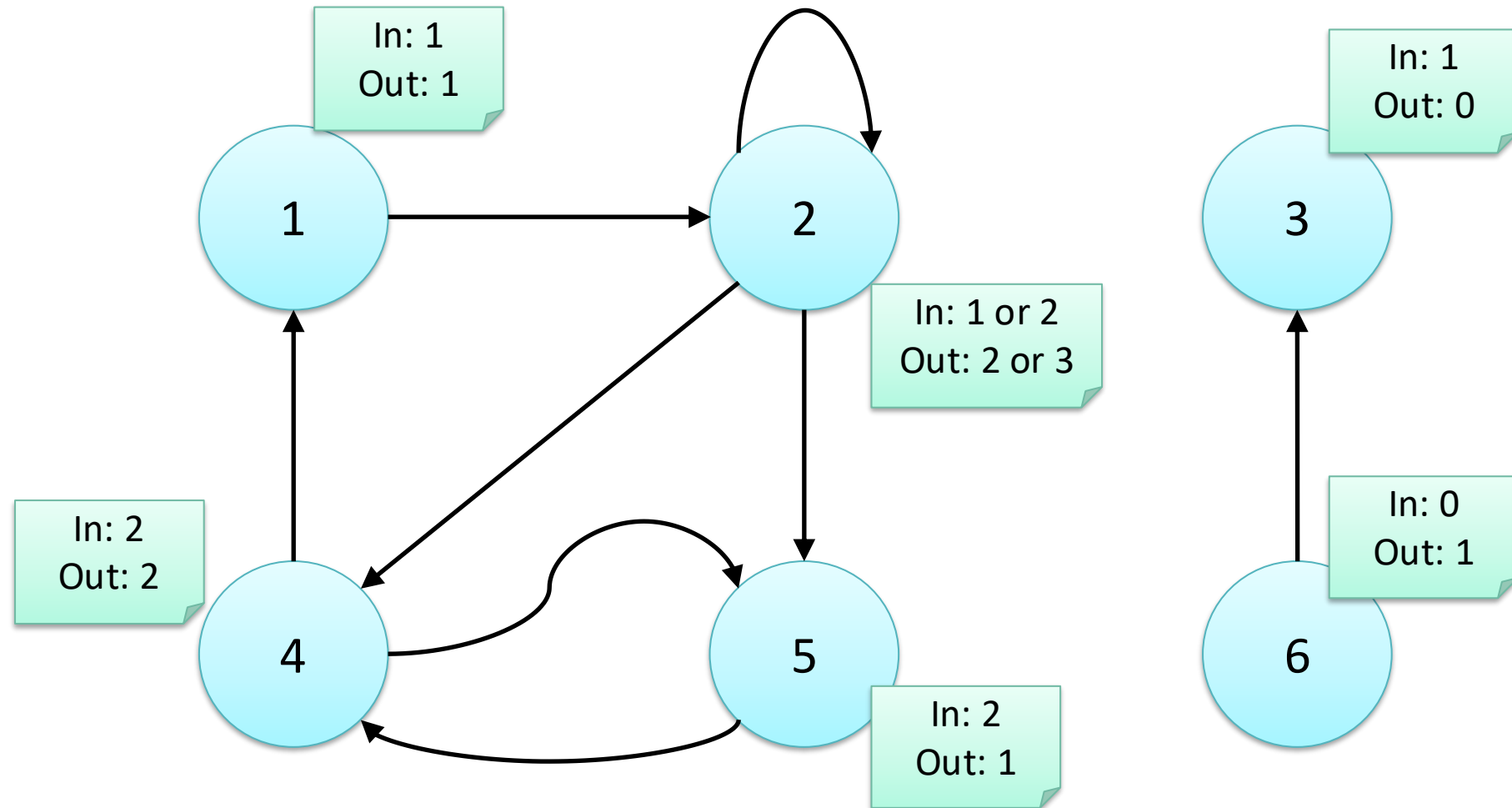
- In an *undirected* graph,
 - the **degree** of a vertex is the number of incident edges
- In a *directed* graph
 - The **in-degree** is the number of incoming edges
 - The **out-degree** is the number of departing edges
 - The **degree** is the sum of in-degree and out-degree
- A vertex with degree 0 is **isolated**



Degree



Degree



Paths

sequenza di nodi ORDINATA, collegate da archi

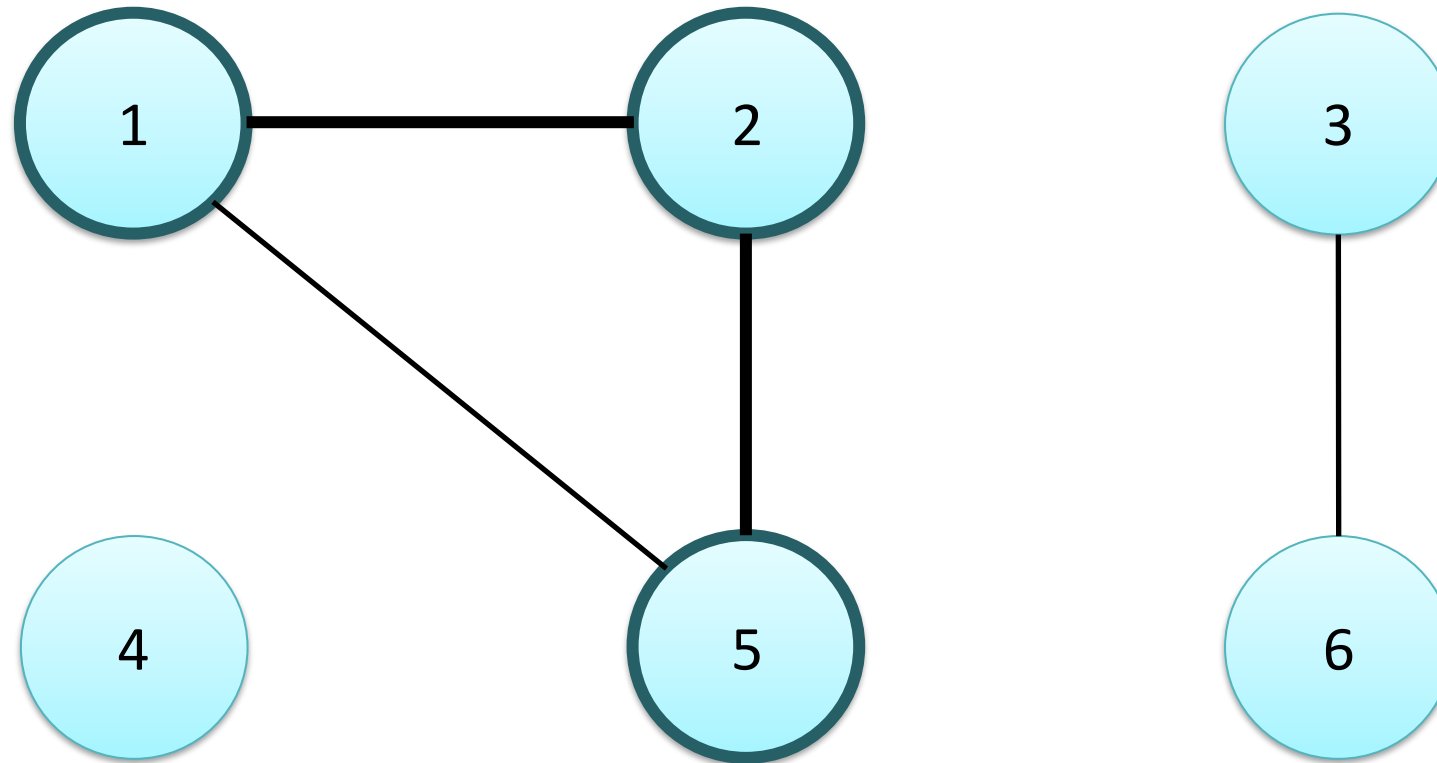
- A **path** on a graph $G=(V,E)$, also called a trail, is a sequence $\{v_1, v_2, \dots, v_n\}$ such that:
 - $v_1, \dots, v_n$ are vertices: $v_i \in V$
 - $(v_1, v_2), (v_2, v_3), \dots, (v_{n-1}, v_n)$ are graph edges: $(v_{i-1}, v_i) \in E$
 - v_i are distinct (for “simple” paths).
- The **length** of a path is the number of edges $(n-1)$
- If there exist a path between v_A and v_B we say that v_B is **reachable** from v_A

Example

Path = (1, 2, 5)

Length = 2

→ ORDINATO



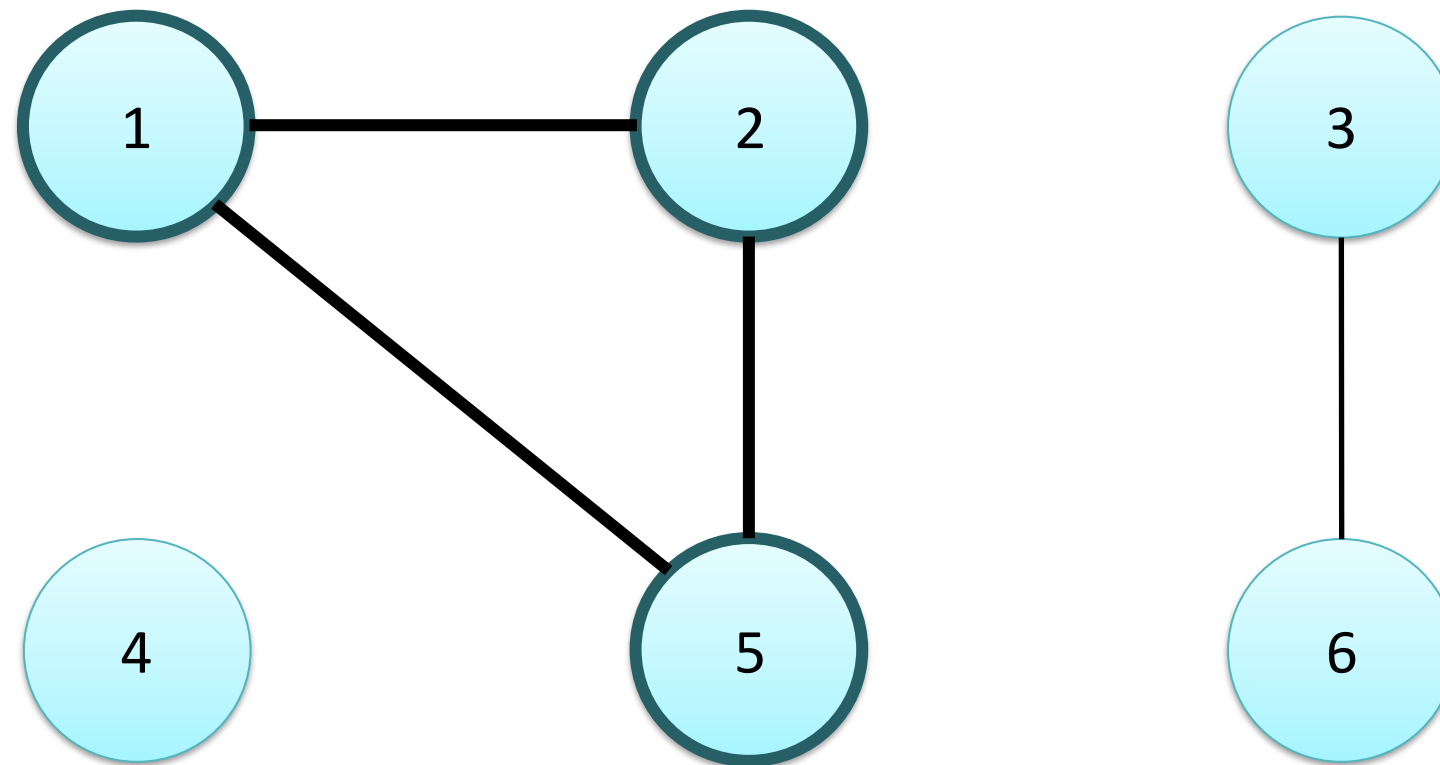
Cycles

- A cycle is a path where $v_1 = v_n$
- A graph with no cycles is said acyclic

NON é possibile
tornare a 1
modo senza ripercorrere
archi già visitati

Example

Path = (1, 2, 5, 1)
Length = 3

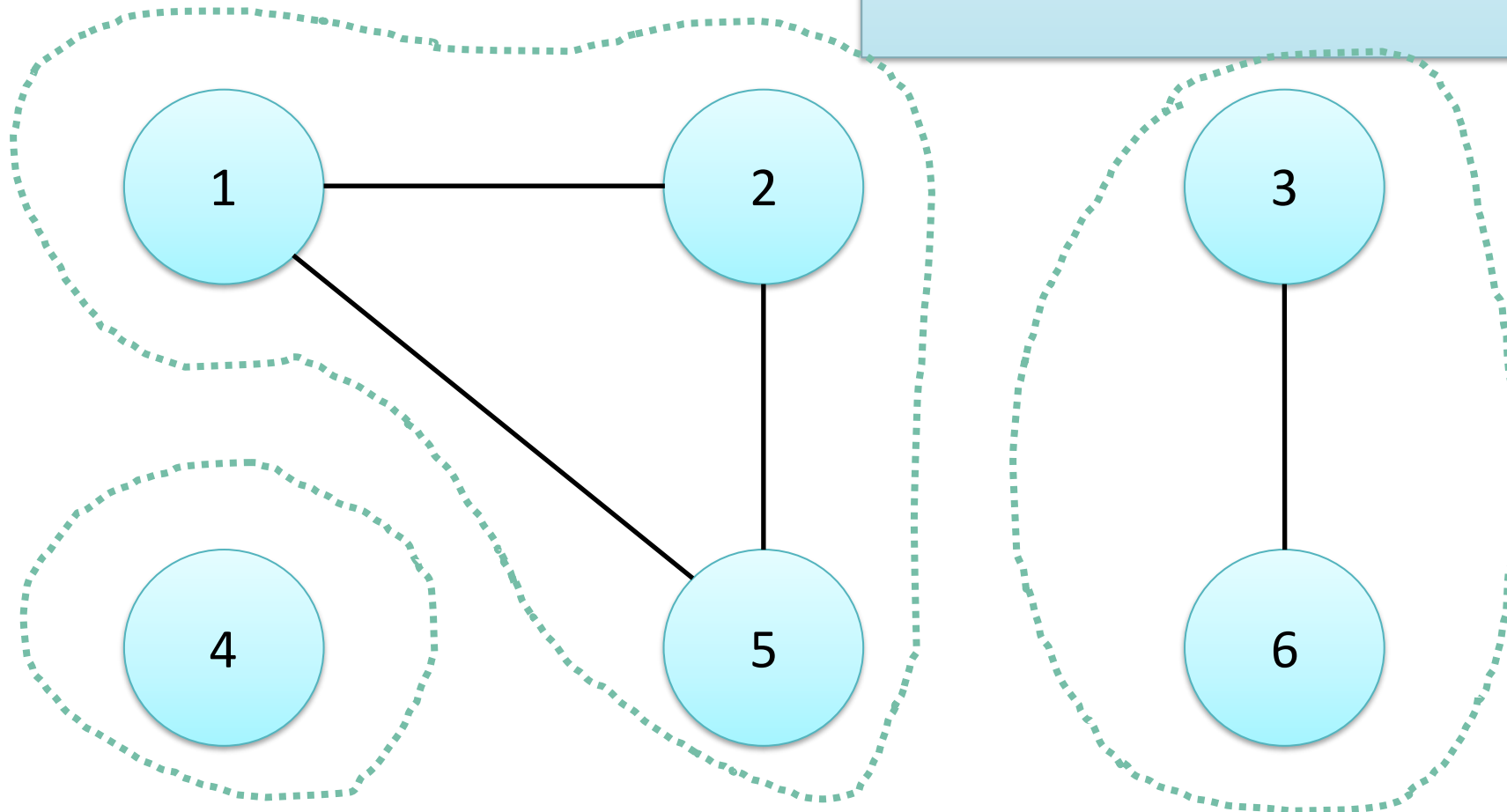


Reachability (Undirected)

- An undirected graph is **connected** if, for every couple of vertices, there is a path connecting them
- The connected sub-graphs of maximum size are called **connected components**
- A connected graph has exactly one connected component

Connected components

The graph is **not** connected.
Connected components = 3
 $\{4\}, \{1, 2, 5\}, \{3, 6\}$

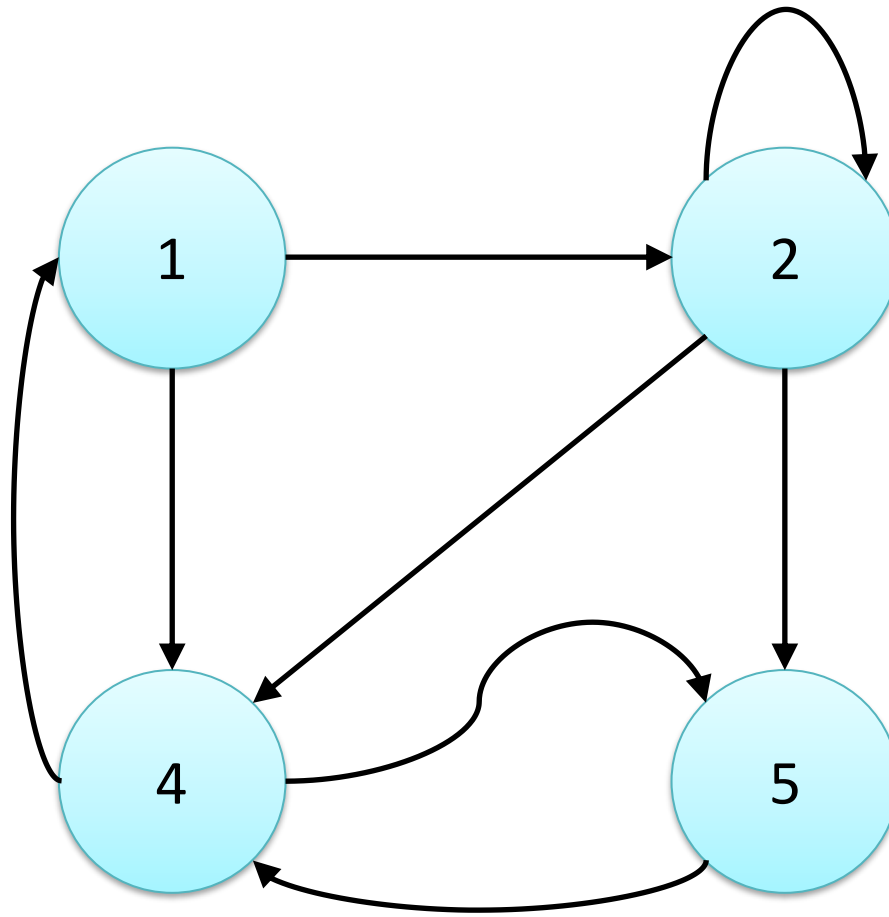


Reachability (Directed)

- A directed graph is **strongly connected** if, for every ordered pair of vertices (v, v') , there exists at least one path connecting v to v'

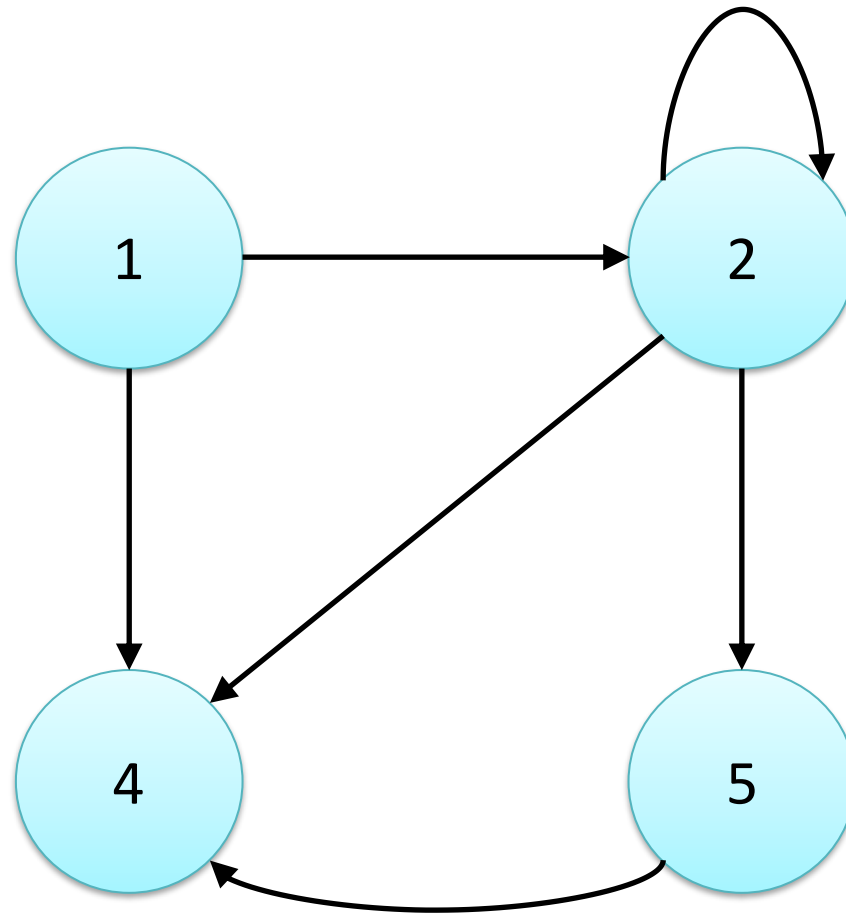
Example

The graph is **strongly connected**



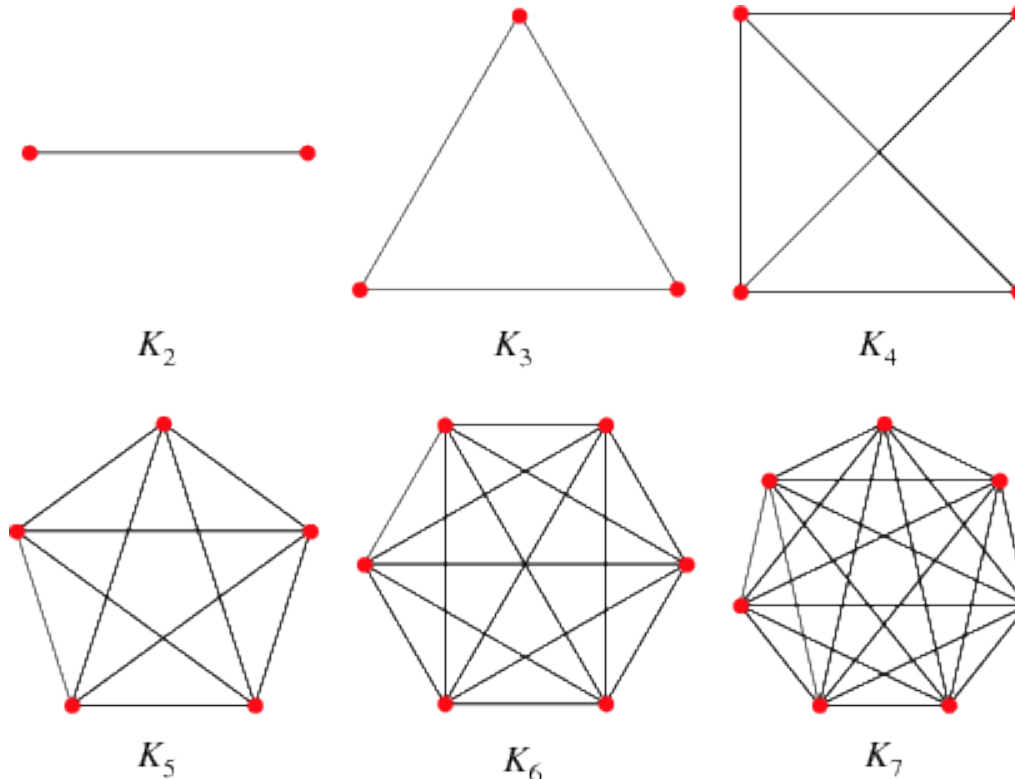
Example

The graph is **not** strongly connected



Complete graph

- A graph is **complete** if, for every pair of vertices, there is an edge connecting them (they are adjacent)
- Symbol: K_n



Complete graph: edges

- In a **complete** graph with n vertices, the number of **edges** is

	Directed	Undirected
No self loops	$n(n - 1)$ $n^2 - n$	$\frac{n(n - 1)}{2}$
With self loops	n^2	$\frac{n(n + 1)}{2}$

Density

- The density of a graph $G=(V,E)$ is the ratio of the number of edges to the total number of possible edges

$$d = \frac{|E(G)|}{|E(K_{|V(G)|})|}$$

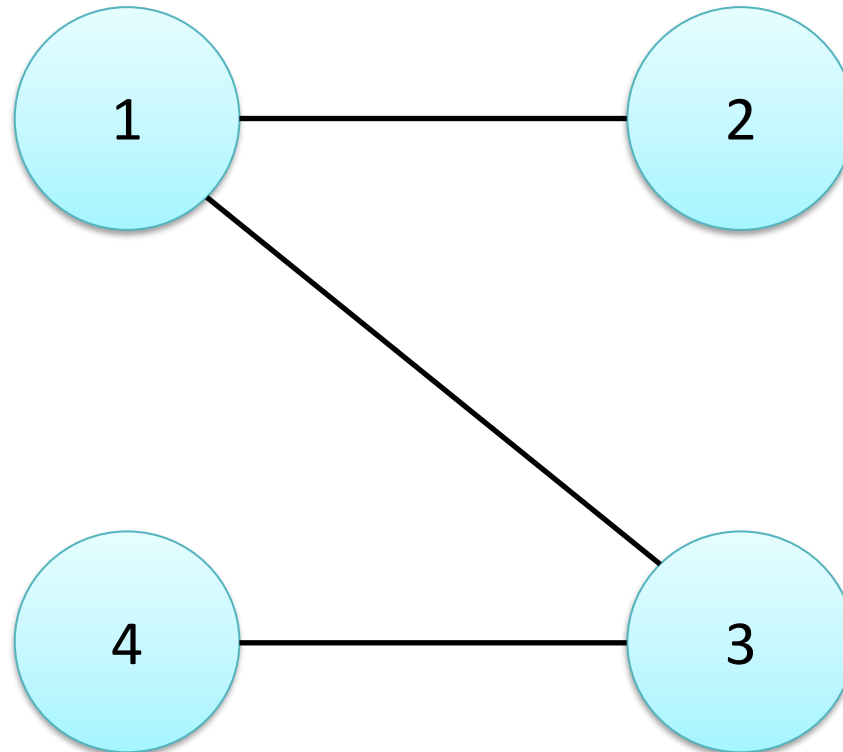
Example

Density = 0.5

$$\frac{3}{6}$$

Existing: 3 edges

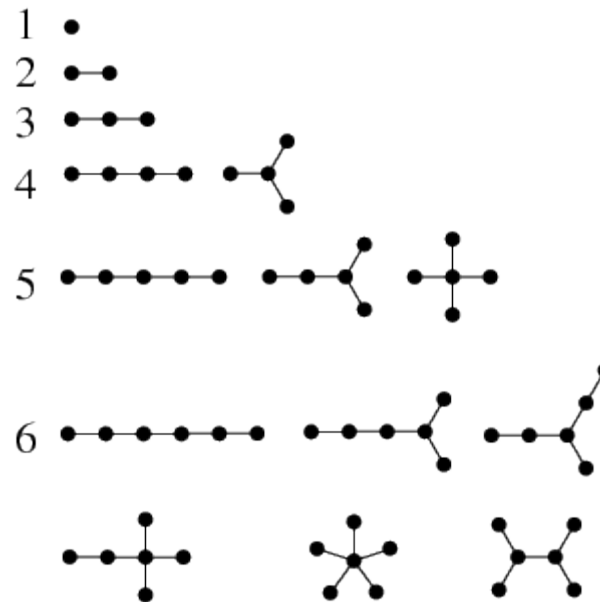
Total: 6 possible edges



Trees and Forests

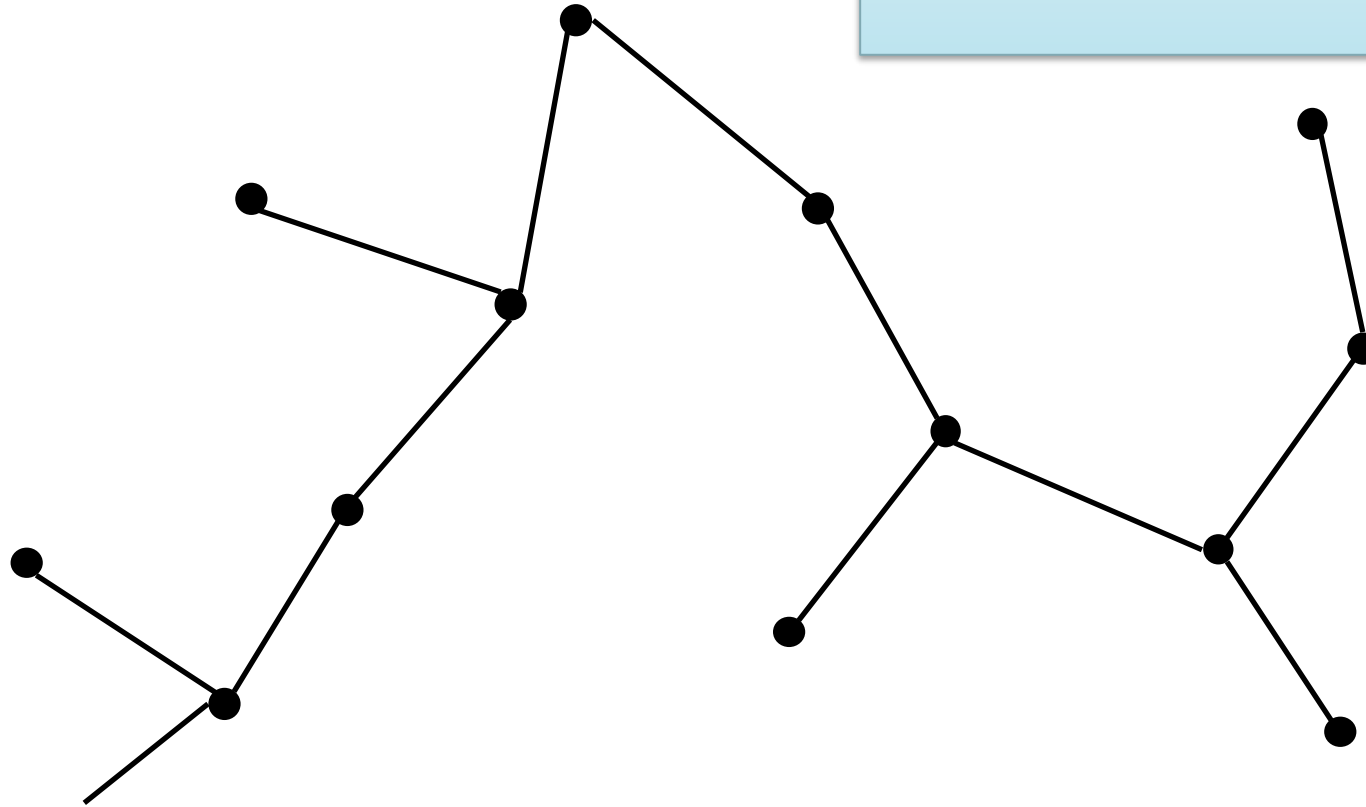
- An undirected acyclic graph is called **forest**
- An undirected acyclic connected graph is called **tree**

*posso raggiungere
ogni altro nodo*

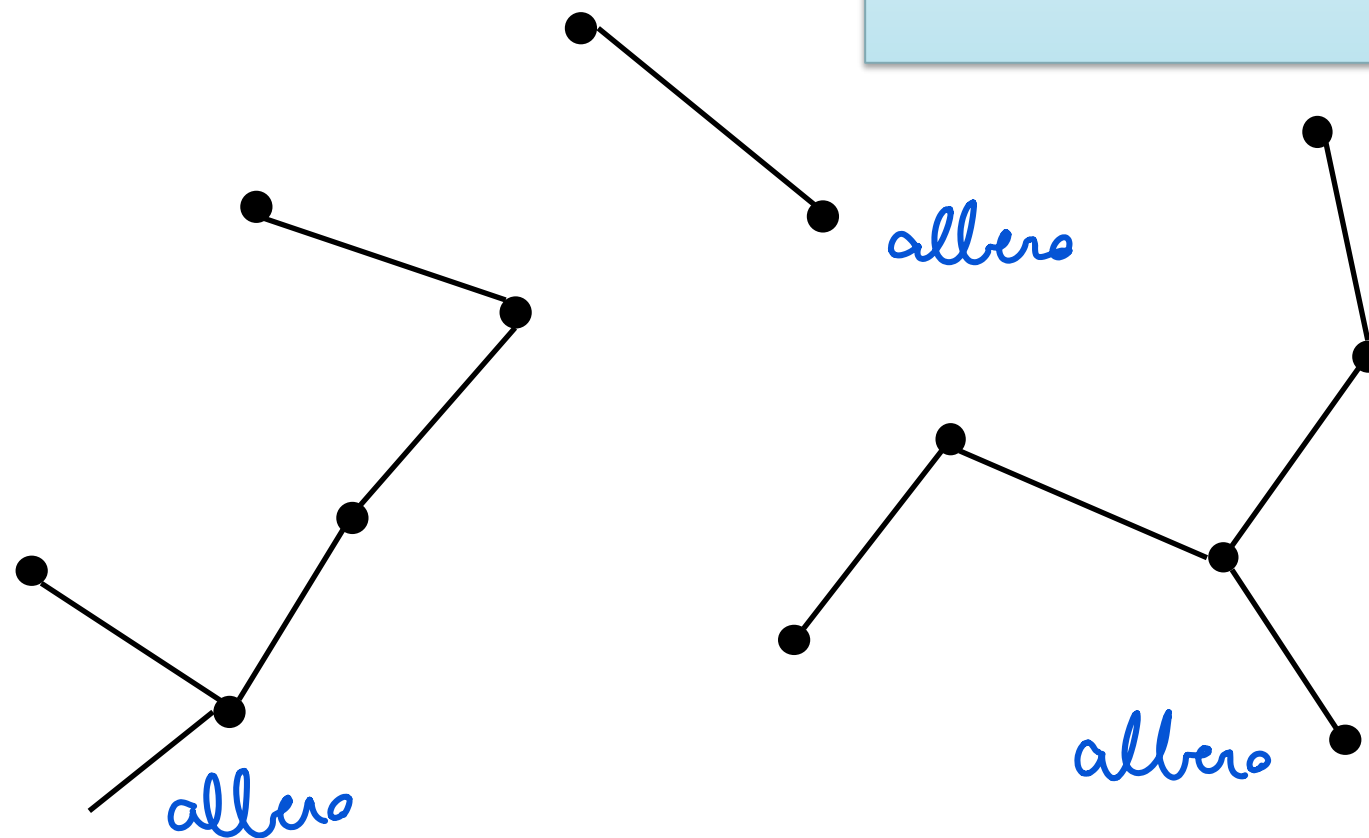


Example

Tree

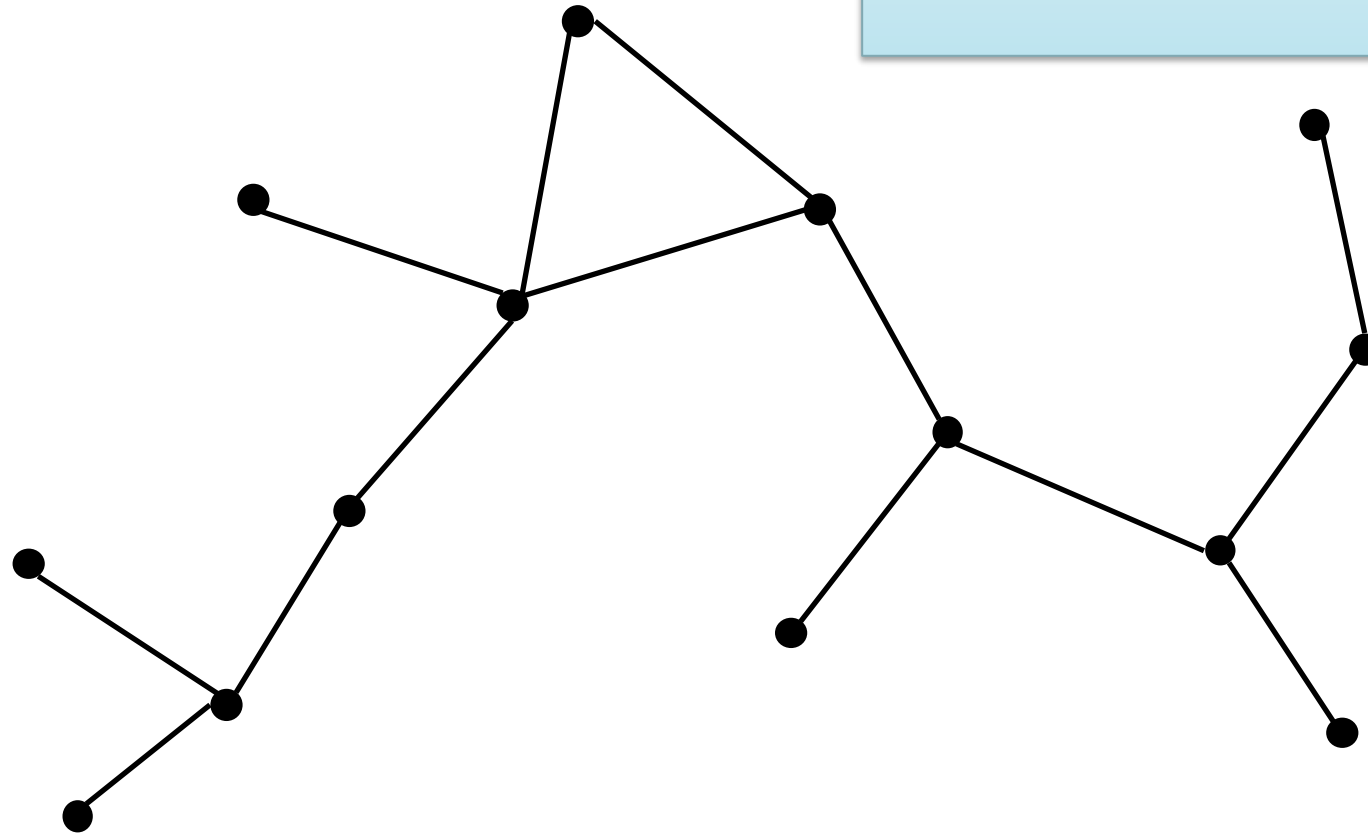


Example



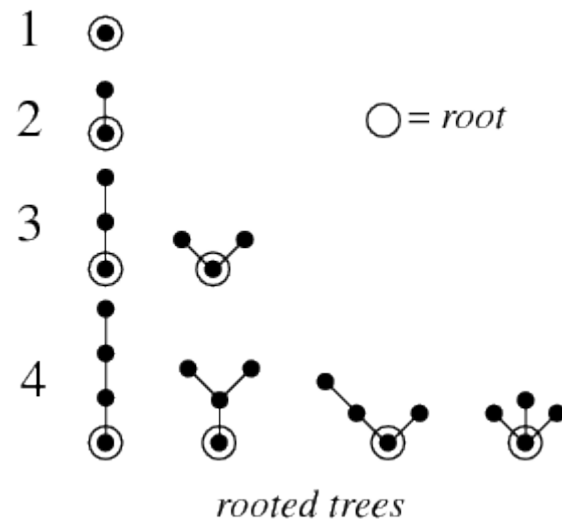
Example

This is not a tree nor a forest
(it contains a cycle)



Rooted trees

- In a tree, a special node may be singled out
- This node is called the “**root**” of the tree
- Any node of a tree can be the root

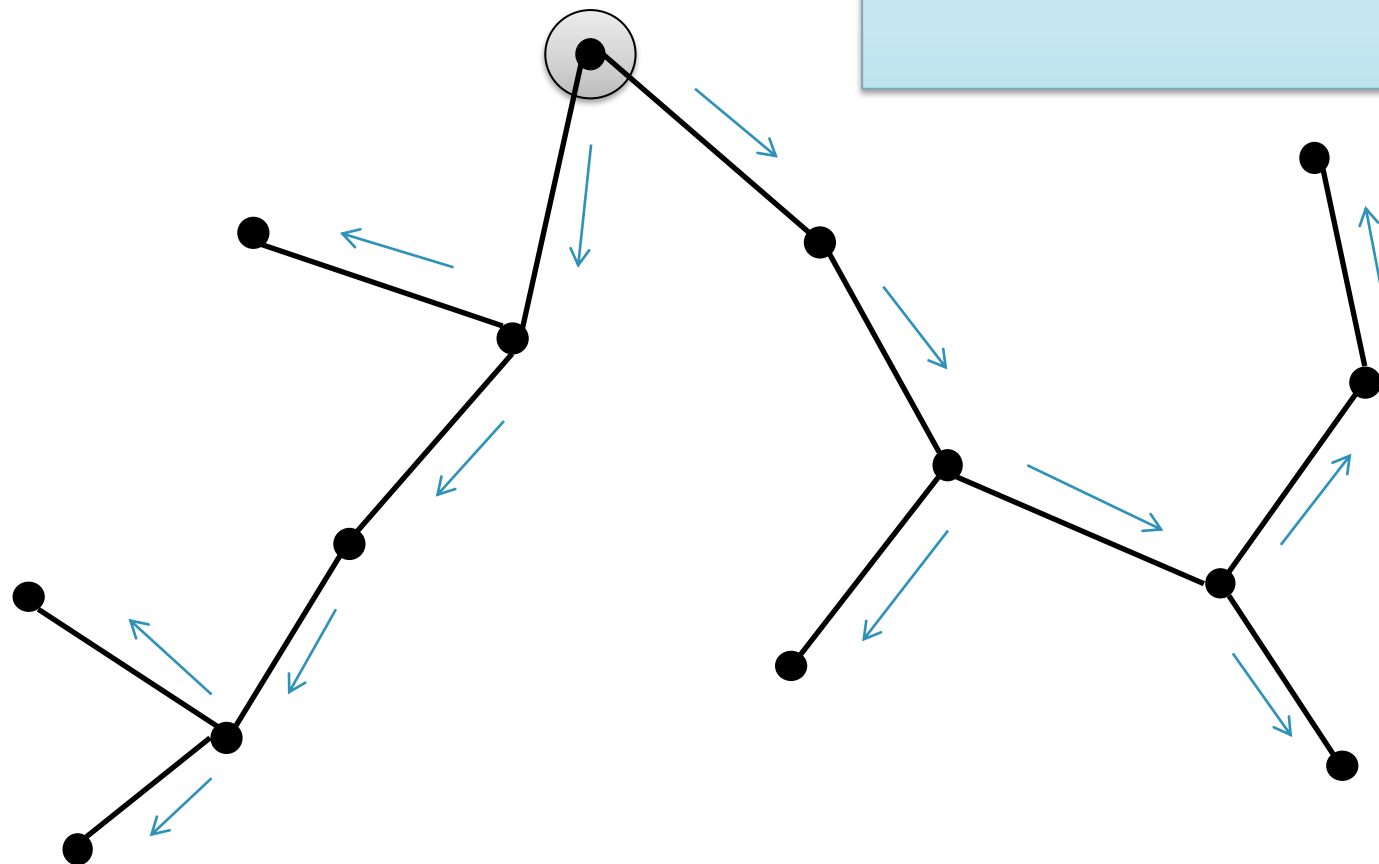


Tree (implicit) ordering

- The root node of a tree **induces an ordering** of the nodes
- The root is the “ancestor” of all other nodes/vertices
 - “children” are “away from the root”
 - “parents” are “towards the root”
- The root is the only node without parents
- All other nodes have exactly one parent
- The furthestmost (children-of-children-of-children...) nodes are “leaves”

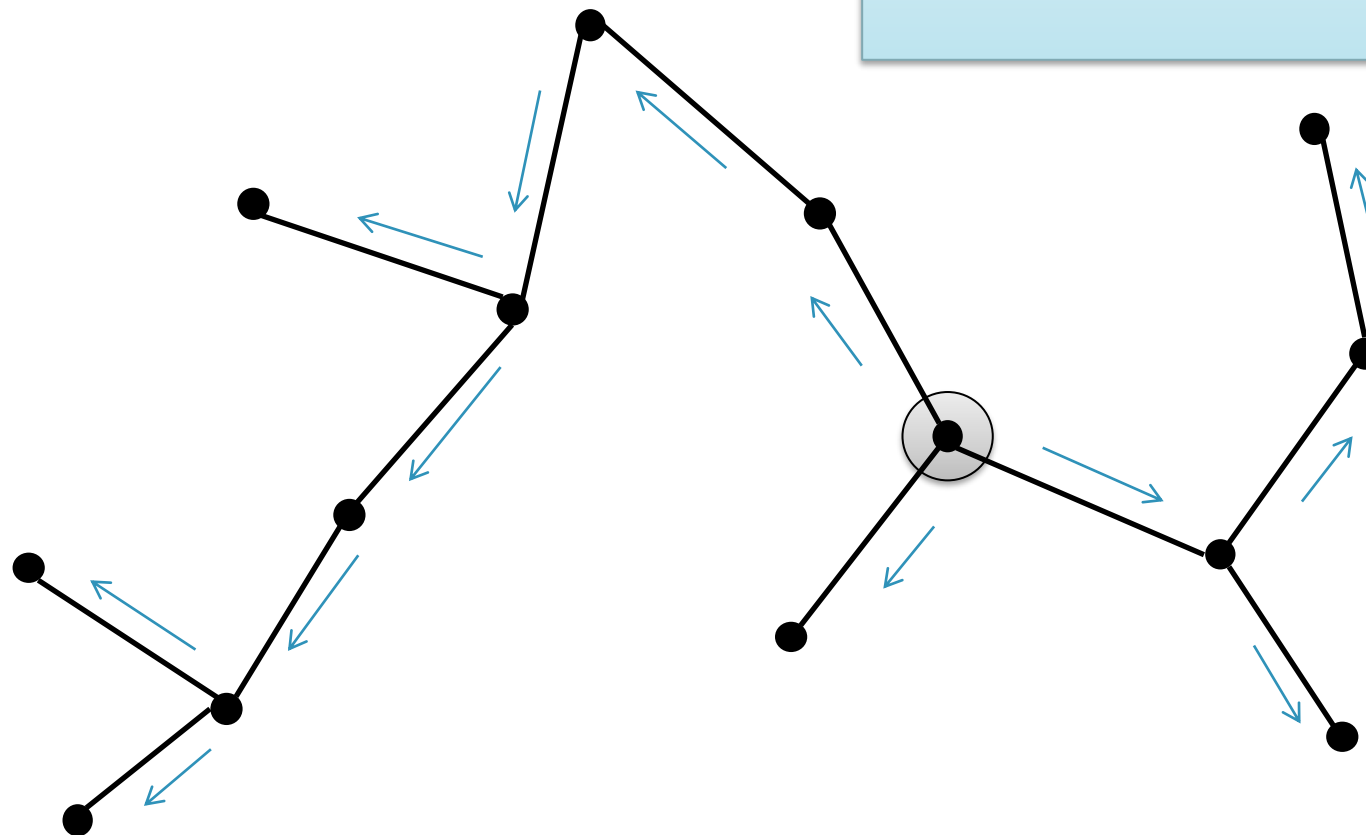
Example

Rooted Tree



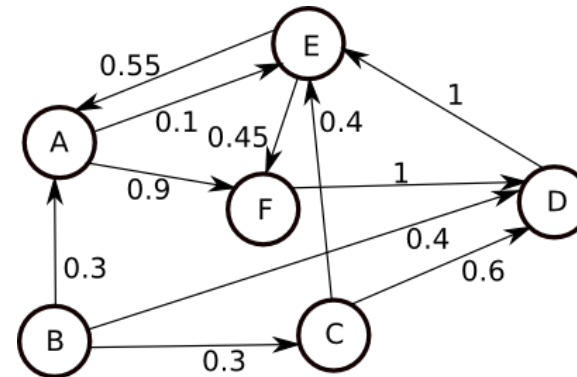
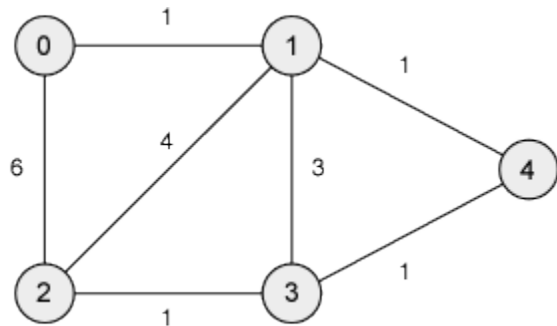
Example

Rooted Tree

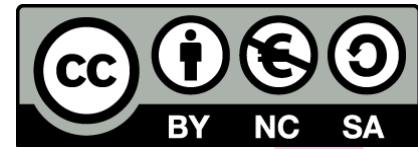


Weighted graphs

- A weighted graph is a graph in which each branch (edge) is given a numerical weight.
- A weighted graph is therefore a special type of labeled graph in which the labels are numbers (which are usually taken to be positive).



License



- These slides are distributed under a Creative Commons license “**Attribution-NonCommercial-ShareAlike 4.0 International (CC BY-NC-SA 4.0)**”
- **You are free to:**
 - **Share** — copy and redistribute the material in any medium or format
 - **Adapt** — remix, transform, and build upon the material
 - The licensor cannot revoke these freedoms as long as you follow the license terms.
- **Under the following terms:**
 - **Attribution** — You must give appropriate credit, provide a link to the license, and indicate if changes were made. You may do so in any reasonable manner, but not in any way that suggests the licensor endorses you or your use.
 - **NonCommercial** — You may not use the material for commercial purposes.
 - **ShareAlike** — If you remix, transform, or build upon the material, you must distribute your contributions under the same license as the original.
 - **No additional restrictions** — You may not apply legal terms or technological measures that legally restrict others from doing anything the license permits.
- <https://creativecommons.org/licenses/by-nc-sa/4.0/>

